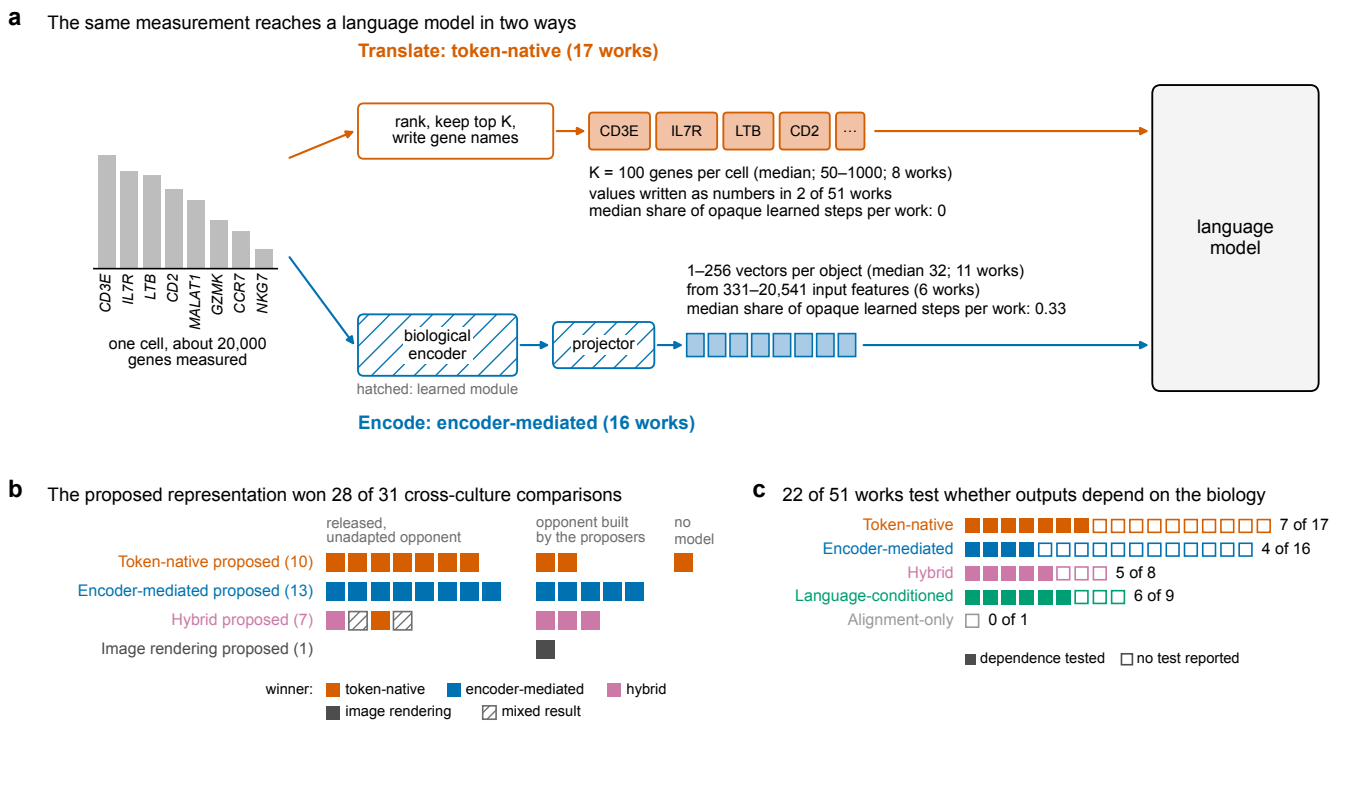

a
The same measurement reaches a language model in two ways
Translate: token-native (17 works)
rank, keep top K, write gene names
CD3E
IL7R
LTB
CD2
···
K = 100 genes per cell (median; 50–1000; 8 works)
values written as numbers in 2 of 51 works
median share of opaque learned steps per work: 0
language model
CD3E
IL7R
LTB
CD2
MALAT1
GZMK
CCR7
NKG7
one cell, about 20,000 genes measured
biological encoder
projector
1–256 vectors per object (median 32; 11 works)
from 331–20,541 input features (6 works)
median share of opaque learned steps per work: 0.33
hatched: learned module
Encode: encoder-mediated (16 works)
b
The proposed representation won 28 of 31 cross-culture comparisons
released, unadapted opponent
opponent built by the proposers
no model
Token-native proposed (10)
Encoder-mediated proposed (13)
Hybrid proposed (7)
Image rendering proposed (1)
winner:
token-native
encoder-mediated
hybrid
image rendering
mixed result
c
22 of 51 works test whether outputs depend on the biology
Token-native
7 of 17
Encoder-mediated
4 of 16
Hybrid
5 of 8
Language-conditioned
6 of 9
Alignment-only
0 of 1
dependence tested
no test reported